Brains vs. Brawn: Machine Learning Models Take on the Octagon

Sricharan Kotala
University of Minnesota - Twin Cities

Abstract

This project applied statistical inference methods to historical data of mixed martial arts (MMA) matches from the Ultimate Fighting Championship (UFC). The goal of the project was to create a model to predict the outcome of Ultimate Fighting Championship matches with the best possible accuracy. The main methods used in the project were Random Forest Classification and Regression and Deep Neural Networks (DNN). The data used for said model was taken from matches between 1996 to 2024. The predictions made by these models were compared against each other. The DNN models achieved up to 83% accuracy in winner prediction and around 77% in method prediction, with overall performance showing competitive results relative to betting odds, with both having comparable predictions to those of the betting sites.

Background

Mixed Martial Arts (MMA) outcomes are notoriously difficult to predict due to the sport's complexity, variability in fighting styles, and the multidimensional nature of athlete performance data. With the growing popularity of UFC events and the high stakes of sports betting markets, there is increasing demand for systems that can analyze historical fight data and generate accurate, interpretable predictions. However, many existing prediction systems rely solely on simple statistical models or human heuristics, which can overlook nonlinear patterns and interactions between features.

The data used for this project was sourced from a publicly available dataset on Kaggle, titled *UFC Complete Dataset (All Events 1996–2024)* [1]. This dataset includes detailed statistics for thousands of mixed martial arts bouts under the Ultimate Fighting Championship (UFC), such as fighter names, striking metrics, grappling statistics, and outcome labels including winner, method of victory, round finished, and time of finish. To ensure model validity and performance consistency, all features were cleaned to remove NaNs and infinities, then normalized for downstream processing.

To address these challenges, we developed a multi-task predictive framework using machine learning to forecast four key fight outcomes: winner, method of victory, round of stoppage, and fight duration. Our system integrates a Random Forest classifier/regressor for baseline performance and a custom Deep Neural Network (DNN) for improved accuracy. The models were trained on curated UFC fight data from 1996 to 2024, using engineered features such as fighter stats, stance, height, weight, fight metrics.

Methodology

1). Random Forest is an ensemble method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. In this project, it was used to predict fight outcomes, victory method, round, and time, serving as a baseline for comparison.

2). Deep Neural Networks (DNNs) are multi-layered models that capture complex patterns in data. They were applied to the same tasks and outperformed Random Forest in several areas due to their ability to model high-dimensional relationships.

Results

Red Fighter	Blue Fighter	Actual Winner	Predicted Winner	Actual Method	Predicted Method	Actual Round	Predicted Round	Actual Time	Predicted Time
Rob Emerson	Rafael Dos Anjos	Rafael Dos Anjos	Rafael Dos Anjos	Decision - Unanimous	Decision - Unanimous	3	3.0	5:00	4:59
Gleison Tibau	Jamie Varner	Gleison Tibau	Jamie Varner	Decision - Split	Decision - Unanimous	3	3.0	5:00	4:50
Rafael Alves	Damir Ismagulov	Damir Ismagulov	Damir Ismagulov	Decision - Unanimous	Decision - Unanimous	3	3.0	5:00	4:17
Benoit Saint Denis	Niklas Stolze	Benoit Saint Denis	Benoit Saint Denis	Submission	Submission	2	2.0	1:32	2:35
Mairbek Taisumov	Tae Hyun Bang	Mairbek Taisumov	Mairbek Taisumov	Decision - Unanimous	Decision - Unanimous	3	3.0	5:00	4:20
Steve Garcia	Charlie Ontiveros	Steve Garcia	Steve Garcia	KO/TKO	KO/TKO	2	2.0	1:51	3:20
Dan Moret	Gilbert Burns	Gilbert Burns	Gilbert Burns	KO/TKO	KO/TKO	2	2.0	0:59	3:07
Thiago Tavares	Matt Wiman	Matt Wiman	Matt Wiman	KO/TKO	Submission	2	3.0	1:57	3:46
Clay Guida	Rafael Dos Anjos	Clay Guida	Clay Guida	Submission	Decision - Unanimous	3	3.0	1:51	4:26
Corey Hill	Joe Veres	Corey Hill	Corey Hill	KO/TKO	KO/TKO	2	2.0	0:37	4:25

Figure 1: Example output of predictions on fighters in the Lightweight division (155 pounds)

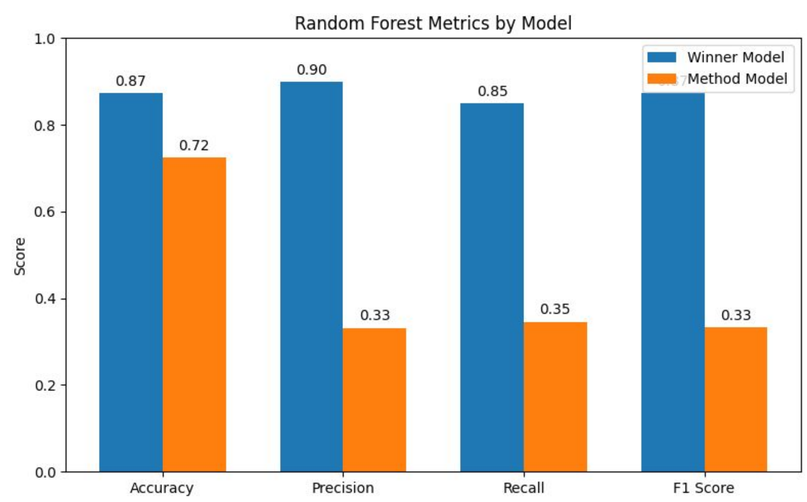

Figure 2: Random Forest Metrics with 150 n_estimators

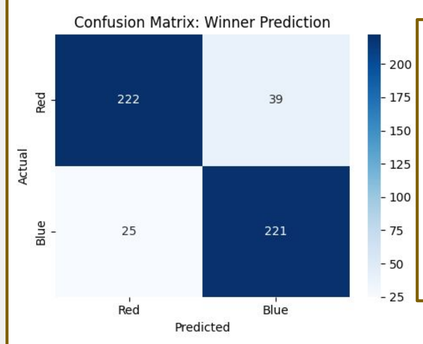

Figure 3: Random Forest Confusion Matrix for Winner

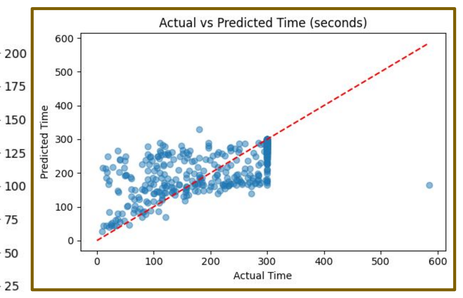

Figure 4: Random Forest Predictions for Finish Time

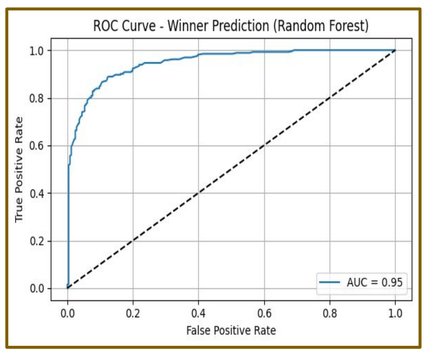

Figure 5: ROC Curve for Random Forest

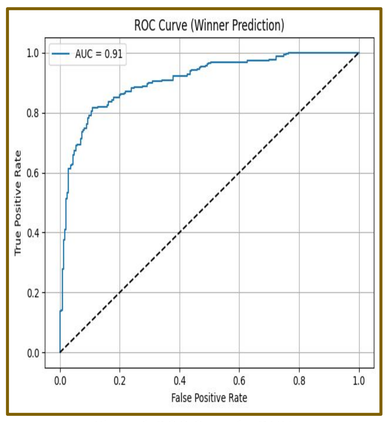

Figure 6: ROC Curve for DNN

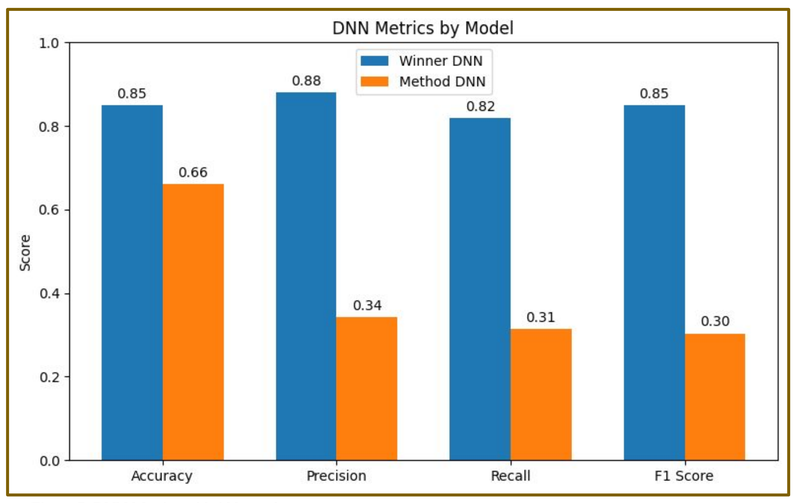

Figure 7: DNN Metrics with 100 epochs

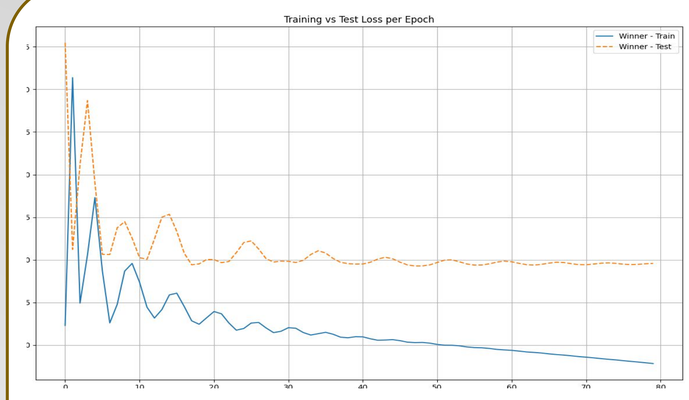

Figure 8: DNN Training and Test Loss per Epoch

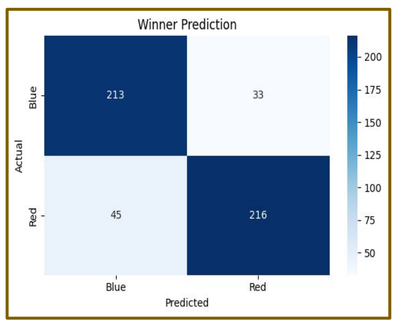

Figure 9: DNN Confusion Matrix for Winner

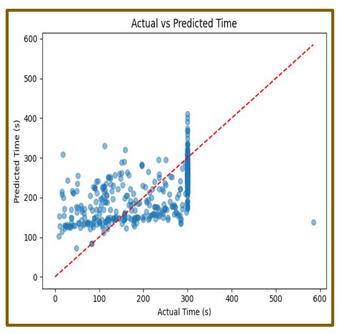

Figure 10: Random Forest Predictions for Finish Time

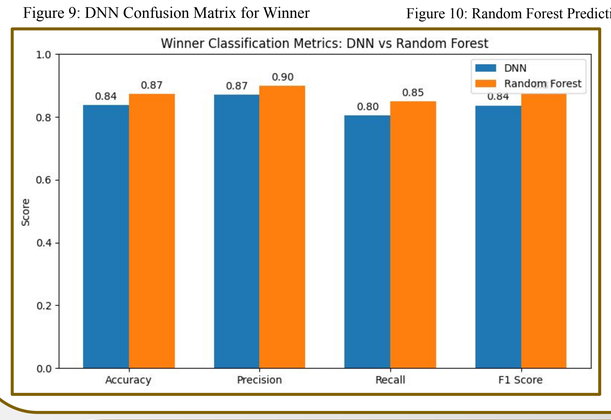

Figure 11: DNN vs Random Forest Metrics

Conclusion & Future Work

Our final system integrates Random Forests and Deep Neural Networks to predict UFC fight outcomes, victory methods, round finishes, and fight durations using historical data. Trained on over two decades of matches, this pipeline balances performance and interpretability across multiple prediction tasks. While DNNs handled complex classification tasks with flexibility, Random Forests consistently outperformed in both accuracy and stability, especially for winner and method predictions. Their ability to reveal feature importance and resist overfitting made them a more reliable choice in this context. Key limitations included class imbalance in rare outcomes and the absence of contextual variables like injuries or weigh-in conditions. Future directions include multi-task architectures, sequential modeling for round-by-round dynamics, incorporation of betting lines, and real-time model updates to improve predictive power and practical deployment.

Acknowledgements

We would like to express our deep gratitude to Professor Shelby Ziccardi (zicca010@umn.edu) for providing us with the necessary skills and tools to create a project of this magnitude. The application of this project was a result of the great teaching and mentoring she has done over the course of the semester. With this support, we were able to create a product that could aid UFC fans in predicting the winner of their favorite fighters to enhance their betting experiences.

[1]: UFC Complete Dataset (All events 1996-2024). (n.d.). Www.kaggle.com. https://www.kaggle.com/datasets/maksbasher/ufc-complete-dataset-all-events-1996-2024